Avi Wadhwa

Ph.D. researcher in theoretical physics
ICTS–TIFR, Bengaluru, India

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Research interests

Post-Newtonian dynamics, effective field theories, AdS/CFT, scattering amplitudes, conformal field theories, cosmology.

Education

2021–present **Ph.D. in Physics**
International Centre for Theoretical Sciences, Tata Institute of Fundamental Research, Bengaluru, India
Thesis topic: Post-Newtonian dynamics and AdS/CFT
Advisor: R. Loganayagam, ICTS, Bengaluru.

2014–2018 **Integrated M.Sc. (Hons.) in Physics**
BITS Pilani – K.K. Birla Goa Campus, India
Thesis: Aspects of Superconformal Representation Theory in Four Dimensions [PDF]
Thesis advisor: Abhijit Gadde, TIFR, Mumbai

Publications and preprints

[1] **Post-Newtonian theory in de Sitter and anti-de Sitter spacetimes**
R. Loganayagam and **Avi Wadhwa**
arXiv:2610.20589 [hep-th] | Preprint, 2026

[2] **Subleading Chern–Simons soft factors in perturbative de Sitter**
Pratik Chattopadhyay and **Avi Wadhwa**
arXiv:2605.05130 [hep-th] | Preprint, 2026

[3] **Soft theorems in Chern–Simons matter theories**
Avi Wadhwa
Physical Review D **113**, 045017 (2026) | arXiv:2509.02542

Research visits

2020 **Tata Institute of Fundamental Research, Mumbai.** String seminar talk

2020 **Indian Institute of Science, Bengaluru.** Visiting student.

2019 **Institute of Physics, Bhubaneswar.** Visiting student.

2018 **Tata Institute of Fundamental Research, Mumbai.** Visiting student for undergraduate thesis.

Fellowships and awards

2021 ICTS Graduate Research Fellowship

CSIR Junior Research Fellowship

2016 **National top 1%**, National Graduate Physics Examination. Merit Certificate.

2014 CBSE Central Sector Scheme of Scholarship for College and University Students

Teaching and academic engagement

2026 **Summer School on Gravitational Wave Astronomy**
Teaching assistant, ICTS–TIFR, Bengaluru. Tutorials for Jan Steinhoff's course, Effective Field Theory Approach to Post-Newtonian Gravity.

2025 **ICTS String Journal Club**
Delivered lectures on conformal field theory and the conformal bootstrap.

2023 **Introduction to General Relativity**
Teaching assistant, ICTS–TIFR, Bengaluru. Tutorials for the graduate course taught by Ashoke Sen.

Conference presentations

2026 **Strings 2026.** Shanghai, China. Poster presentation.

Pre-Strings School. SIMIS, Shanghai, China. Poster presentation.

2025 **Indian Strings Meeting.** NISER and IIT Bhubaneswar, India. Poster presentation.

28th CAPRA Meeting on Radiation Reaction in General Relativity. University of Southampton, United Kingdom. Flash talk.

Schools, programs and conferences attended

2026 Summer School on Gravitational Wave Astronomy. ICTS–TIFR, Bengaluru.

School for Advanced Topics in Particle Physics: QCD, Effective Field Theories, and Nuclear Physics. ICTS–TIFR, Bengaluru.

2025 From Asymptotic Symmetries to Flat Holography: Theoretical Aspects and Observable Consequences. INFN Galileo Galilei Institute, Florence, Italy.

Loop-the-Loop-2: Feynman calculus and its applications to gravity and particle physics. Online.

A Hundred Years of Quantum Mechanics. ICTS–TIFR, Bengaluru.

Strings 2025, Annual Meeting. Online.

2024 Hearing beyond the Standard Model with Cosmic Sources of Gravitational Waves. ICTS–TIFR, Bengaluru.

Quantum Information, Quantum Field Theory, and Gravity. ICTS–TIFR, Bengaluru.

2023 Field Theory and Turbulence. ICTS–TIFR, Bengaluru.

Indian Strings Meeting. IIT Bombay, India.

2022 Non-perturbative and Numerical Approaches to Quantum Gravity, String Theory and Holography. ICTS–TIFR, Bengaluru.

IFCAM Summer School on Mathematical Aspects of Quantum Mechanics. IFCAM, Indian Institute of Science, Bengaluru.

2021 Physics of the Early Universe. ICTS–TIFR, Bengaluru.